ECE408/CS483/CSE408 Fall 2024

Applied Parallel Programming

Lecture 20: Data Transfer and CUDA Streams (Task Parallelism)

© David Kirk/NVIDIA and Wen-mei W. Hwu, 2007-2018
ECE408/CS483/ University of Illinois at Urbana-Champaign

Course Reminders

- PM2 is due this Friday
- Lab 8 is due next week
- PM3 should be out after PM2 is submitted
  - This is a much more involved PM, start early
  - Do not wait to submit until last minute!
  - Competition description will be out soon, but first you need to finish PM3. So, do not yet work on the competition part.

https://wiki.illinois.edu/wiki/display/ECE408/Class+Schedule

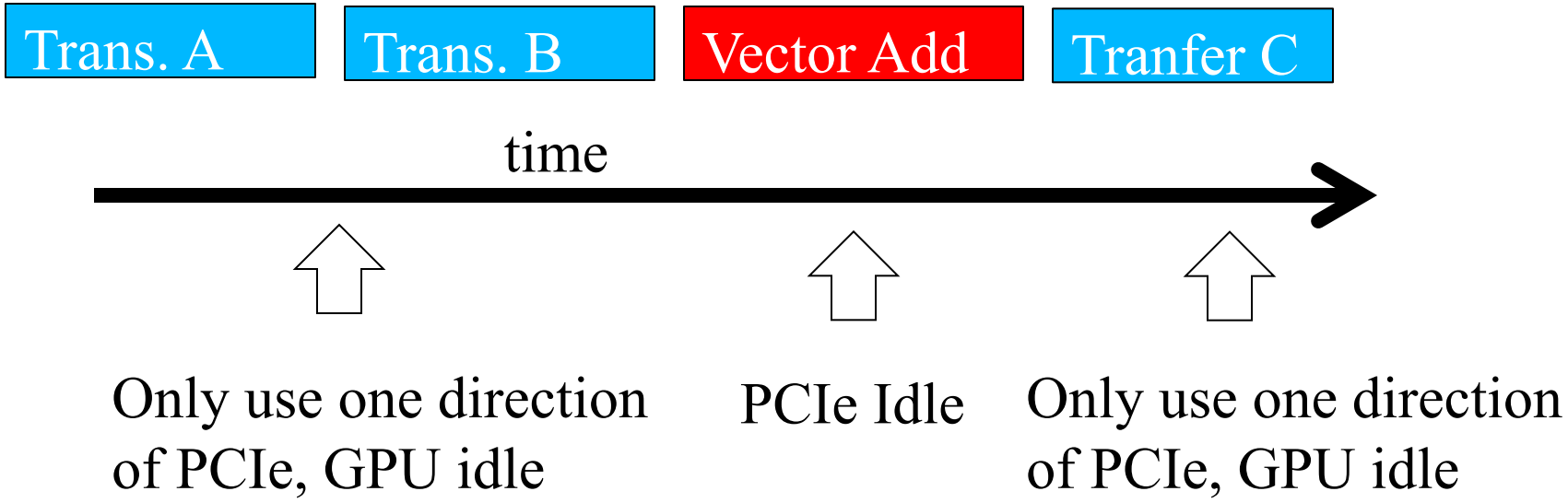
Objective

- To learn more advanced features of the CUDA APIs for data transfer and kernel launch
  - Task parallelism for overlapping data transfer with kernel computation
  - CUDA streams

© David Kirk/NVIDIA and Wen-mei W. Hwu, 2007-2018
ECE408/CS483/ University of Illinois at Urbana-Champaign

Serialized Data Transfer

- So far, the way we use `cudaMemcpy` serializes data transfer and GPU computation



© David Kirk/NVIDIA and Wen-mei W. Hwu, 2007-2018
ECE408/CS483/ University of Illinois at Urbana-Champaign

Device Overlap

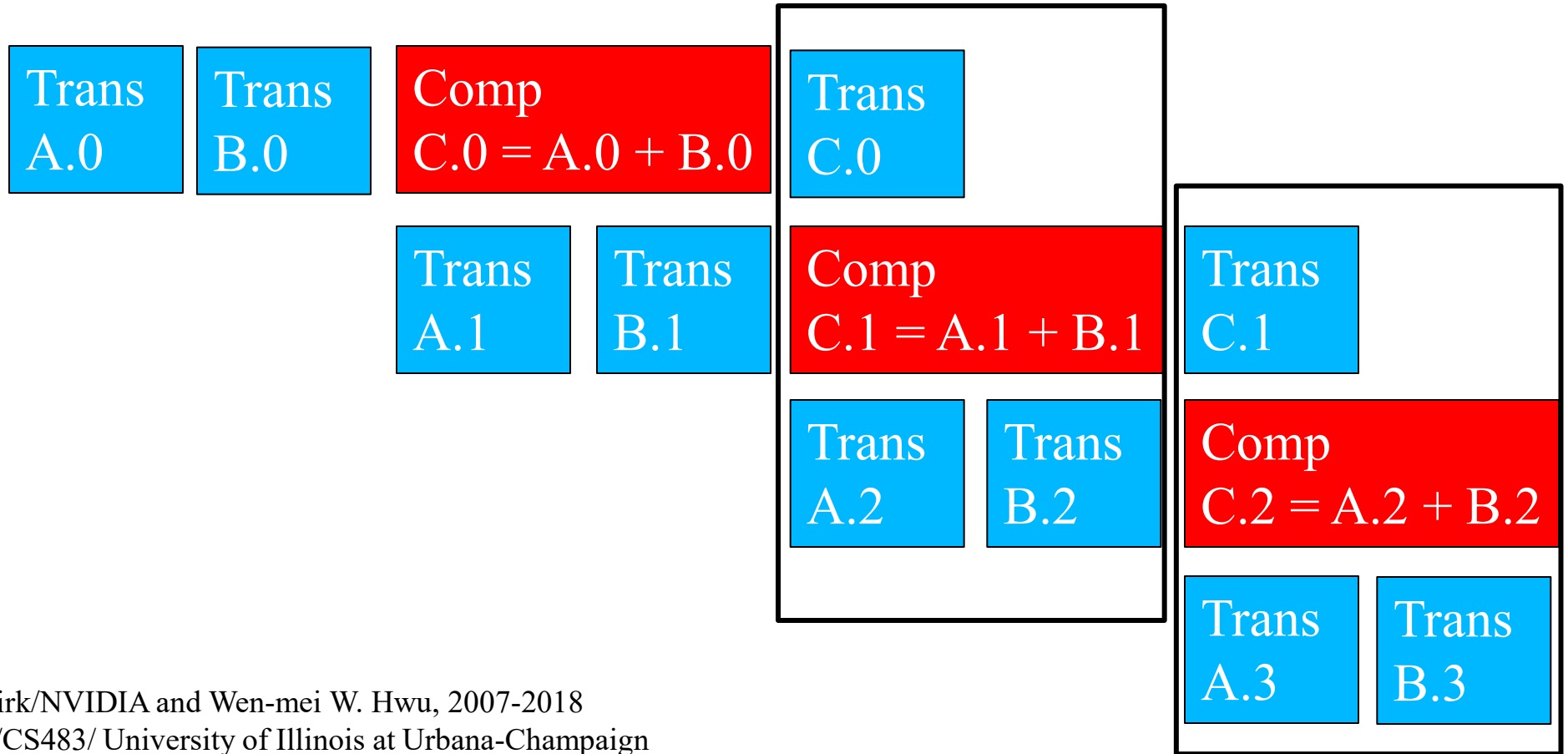
- Most CUDA devices support *device overlap*
  - *Simultaneously execute a kernel while performing a copy between device and host memory*

```
int dev_count;
cudaDeviceProp prop;

cudaGetDeviceCount( &dev_count);
for (int i = 0; i < dev_count; i++) {
  cudaGetDeviceProperties(&prop, i);

  if (prop.deviceOverlap) …
```

© David Kirk/NVIDIA and Wen-mei W. Hwu, 2007-2018
ECE408/CS483/ University of Illinois at Urbana-Champaign

Overlapped (Pipelined) Timing

- Divide large vectors into segments
- Overlap transfer and compute of adjacent segments



© David Kirk/NVIDIA and Wen-mei W. Hwu, 2007-2018
ECE408/CS483/ University of Illinois at Urbana-Champaign

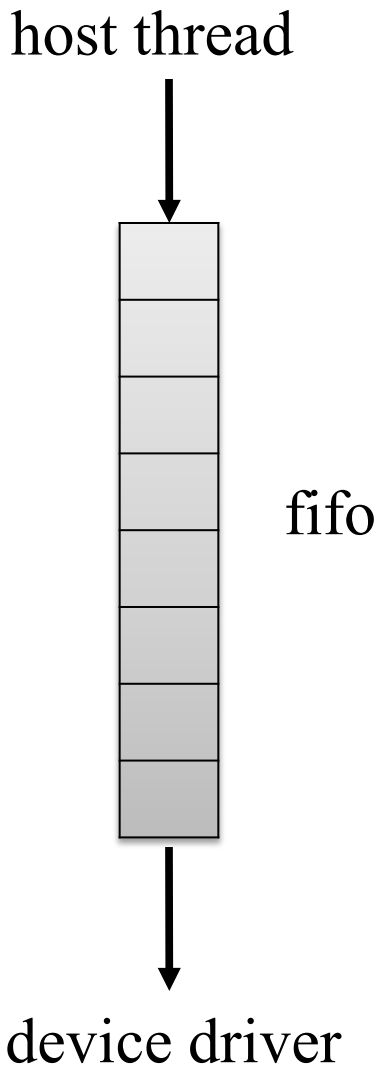
Using CUDA Streams and Asynchronous Memcpy

- CUDA supports parallel execution of kernels and cudaMemcpy with **streams**
- Each stream **is a queue of operations** (kernel launches and cudaMemcpy's)
- Operations (tasks) in different streams
  - can execute in parallel
  - a version of **task parallelism**

© David Kirk/NVIDIA and Wen-mei W. Hwu, 2007-2018
ECE408/CS483/ University of Illinois at Urbana-Champaign

Streams

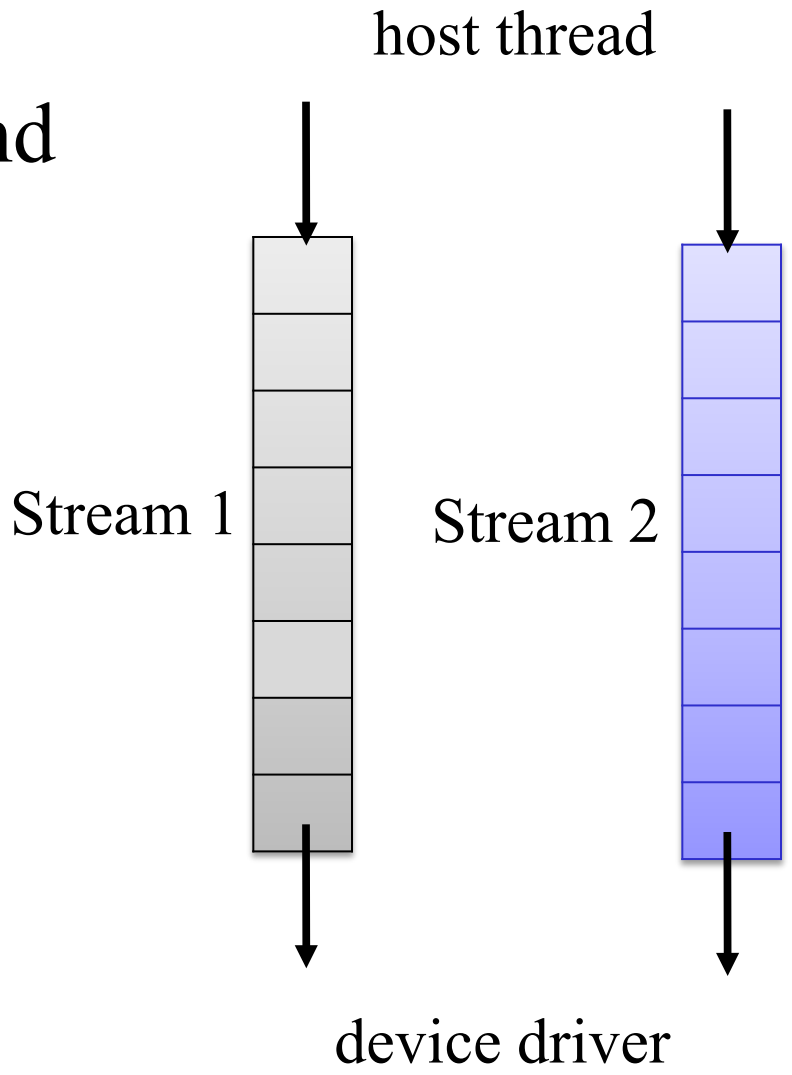
- Device requests made from the host code are put into a queue
  - Queue processed asynchronously by the driver and device. Called a "Stream".
  - Driver ensures that commands in the queue are processed strictly in sequence. Memory copies end before kernel launch, etc.



© David Kirk/NVIDIA and Wen-mei W. Hwu, 2007-2018
ECE408/CS483/ University of Illinois at Urbana-Champaign

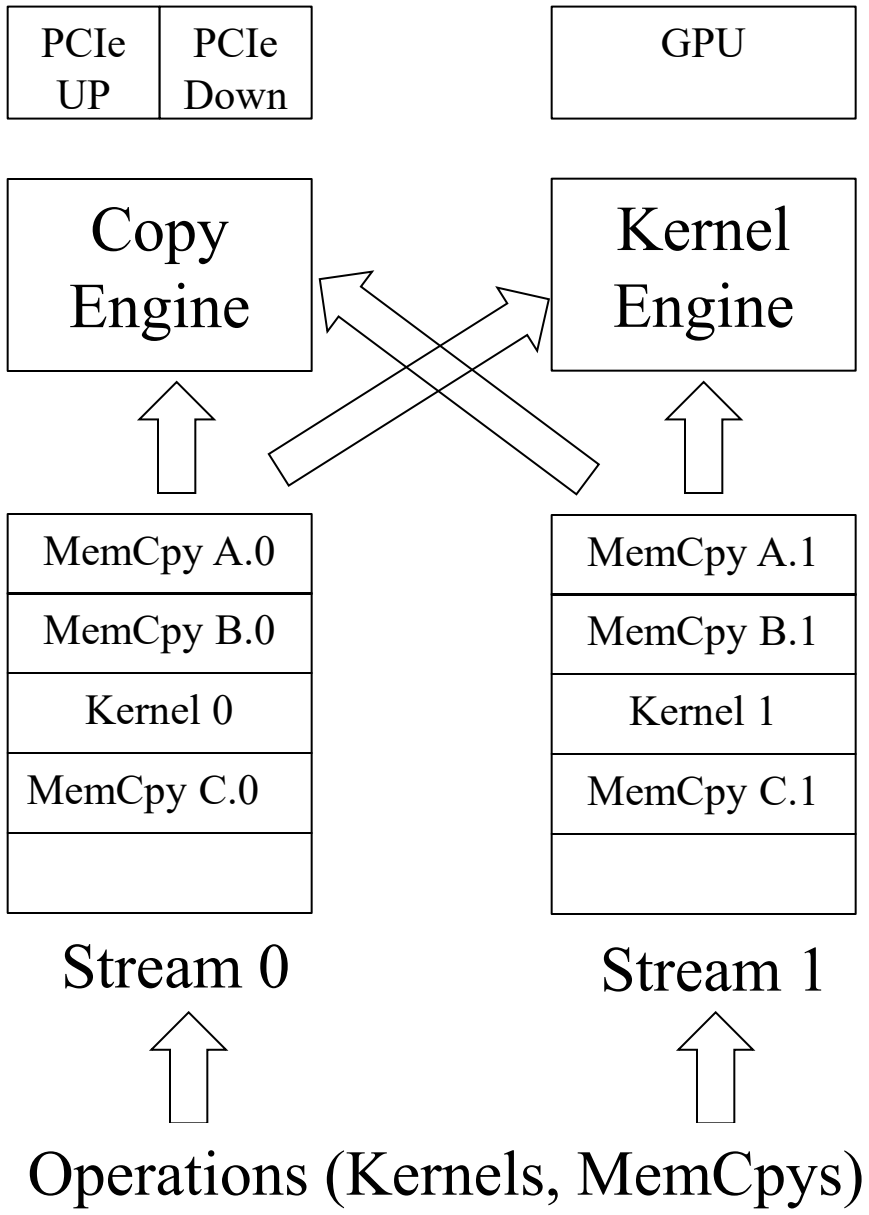
Streams cont.

- To allow concurrent copying and kernel execution, multiple queues are required



© David Kirk/NVIDIA and Wen-mei W. Hwu, 2007-2018
ECE408/CS483/ University of Illinois at Urbana-Champaign

Conceptual View of Streams



© David Kirk/NVIDIA and Wen-mei W. Hwu, 2007-2018
ECE408/CS483/ University of Illinois at Urbana-Champaign

A Simple Multi-Stream Host Code

```
cudaStream_t      stream0, stream1;
cudaStreamCreate(&stream0);
cudaStreamCreate(&stream1);
float *d_A0, *d_B0, *d_C0;    // device memory for stream 0
float *d_A1, *d_B1, *d_C1;    // device memory for stream 1

// cudaMalloc for d_A0, d_B0, d_C0, d_A1, d_B1, d_C1 go here

for (int i=0; i<n; i+=SegSize*2) {
  // copy data in stream0
  // lunch kernel in stream0
  // copy results in stream0
  // copy data in stream1
  // lunch kernel in stream1
  // copy results in stream1
}
```

© David Kirk/NVIDIA and Wen-mei W. Hwu, 2007-2018
ECE408/CS483/ University of Illinois at Urbana-Champaign

A Simple Multi-Stream Host Code

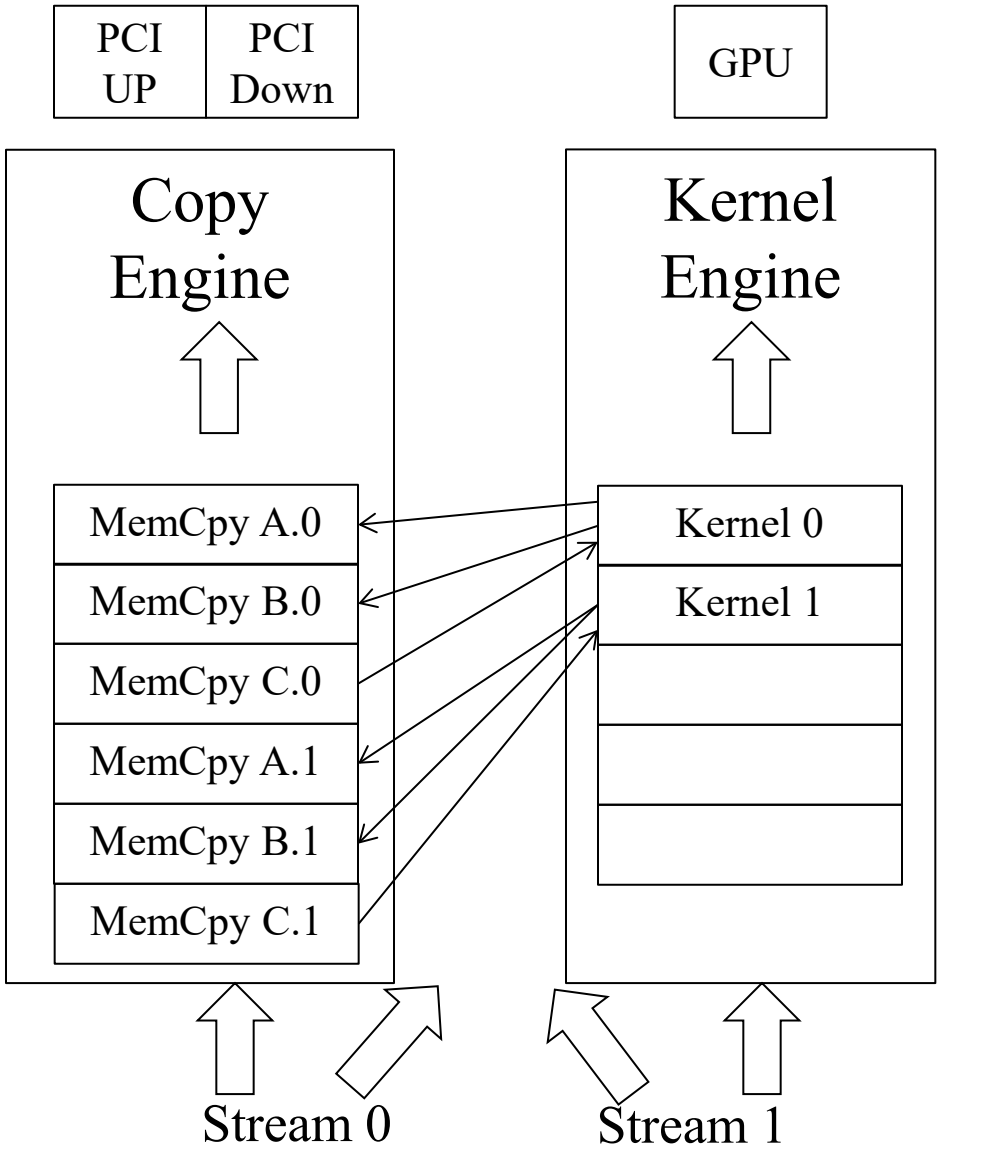
```
for (int i=0; i<n; i+=SegSize*2)
{
  cudaMemcpyAsync(d_A0, h_A+i, SegSize*sizeof(float),.., stream0);
  cudaMemcpyAsync(d_B0, h_B+i, SegSize*sizeof(float),.., stream0);
  vecAdd<<<SegSize/256, 256, 0, stream0>>>(d_A0, d_B0, …);
  cudaMemcpyAsync(h_C+i, d_C0, SegSize*sizeof(float),.., stream0);

  cudaMemcpyAsync(d_A1, h_A+i+SegSize;
                                SegSize*sizeof(float),.., stream1);
  cudaMemcpyAsync(d_B1, h_B+i+SegSize;
                                SegSize*sizeof(float),.., stream1);
  vecAdd<<<SegSize/256, 256, 0, stream1>>>(d_A1, d_B1, …);
  cudaMemcpyAsync(h_C+i+SegSize, d_C1,
                                SegSize*sizeof(float),.., stream1);
}
```


© David Kirk/NVIDIA and Wen-mei W. Hwu, 2007-2018
ECE408/CS483/ University of Illinois at Urbana-Champaign

Older GPUs Support Streams in Software

Task at head of queue waits for dependencies (arcs) before executing.

For example, Kernel 0 waits for MemCpy A.0 and MemCpy B.0 to finish.

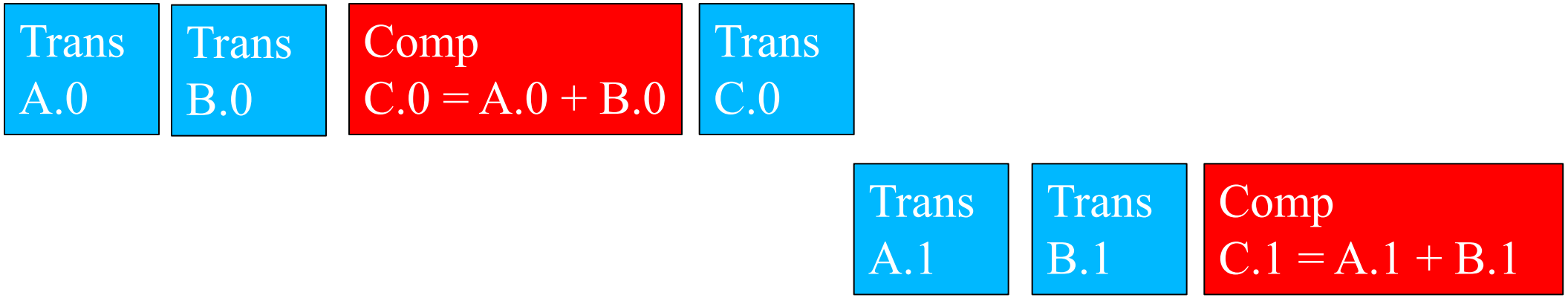


Operations (Kernels, MemCpys)

© David Kirk/NVIDIA and Wen-mei W. Hwu, 2007-2018
ECE408/CS483/ University of Illinois at Urbana-Champaign

Not quite the overlap we want

- C.0 blocks A.1 and B.1 in the copy engine queue



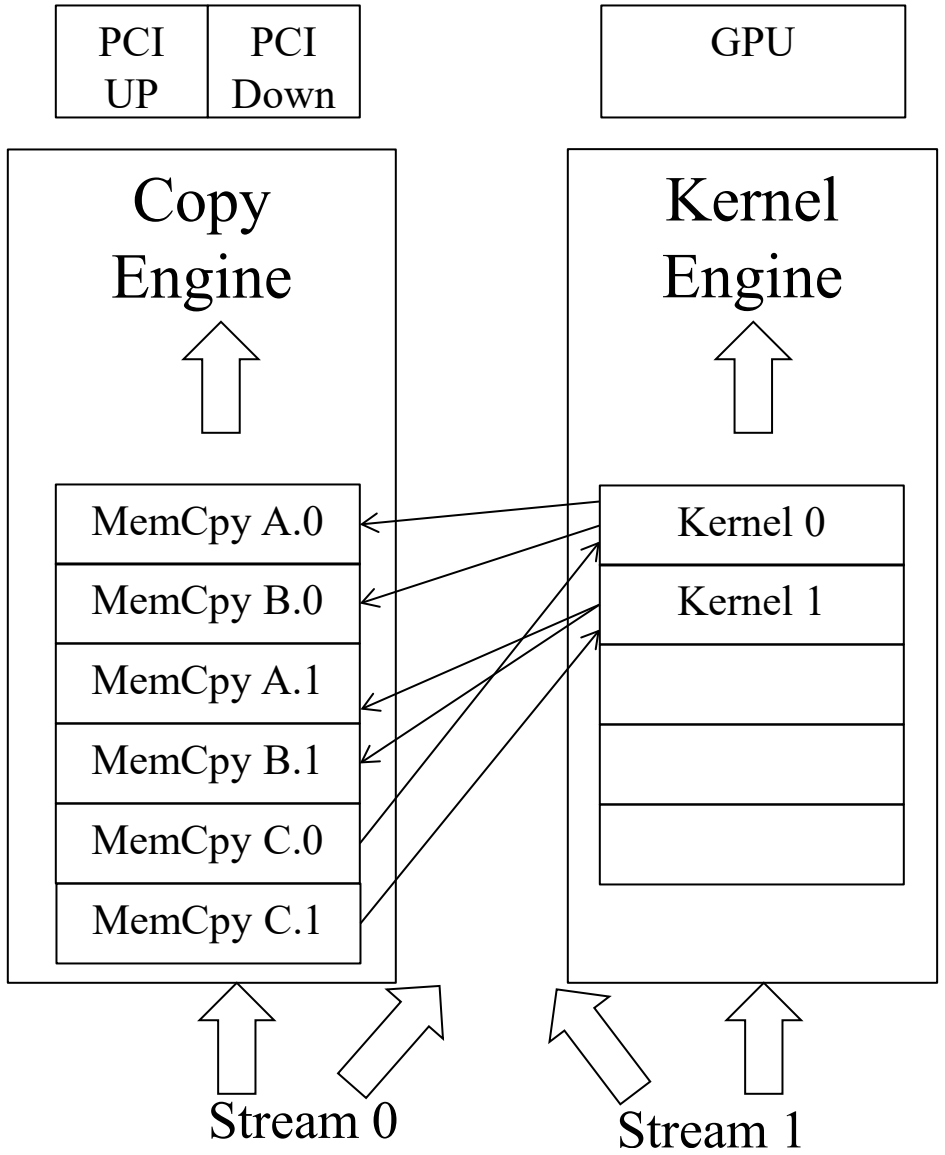
© David Kirk/NVIDIA and Wen-mei W. Hwu, 2007-2018
ECE408/CS483/ University of Illinois at Urbana-Champaign

A Better Multi-Stream Host Code

```
for (int i=0; i<n; i+=SegSize*2) {

  cudaMemCpyAsync(d_A0, h_A+i; SegSize*sizeof(float),.., stream0);
  cudaMemCpyAsync(d_B0, h_B+i; SegSize*sizeof(float),.., stream0);
  cudaMemCpyAsync(d_A1, h_A+i+SegSize;
                                SegSize*sizeof(float),.., stream1);
  cudaMemCpyAsync(d_B1, h_B+i+SegSize;
                                SegSize*sizeof(float),.., stream1);

  vecAdd<<<SegSize/256, 256, 0, stream0>>>(d_A0, d_B0, …);
  vecAdd<<<SegSize/256, 256, 0, stream1>>>(d_A1, d_B1, …);
  cudaMemCpyAsync(d_C0, h_C+I; SegSize*sizeof(float),.., stream0);
  cudaMemCpyAsync(d_C1, h_C+i+SegSize;
                                SegSize*sizeof(float),.., stream1);
}
```


© David Kirk/NVIDIA and Wen-mei W. Hwu, 2007-2018
ECE408/CS483/ University of Illinois at Urbana-Champaign

A View Closer to Reality

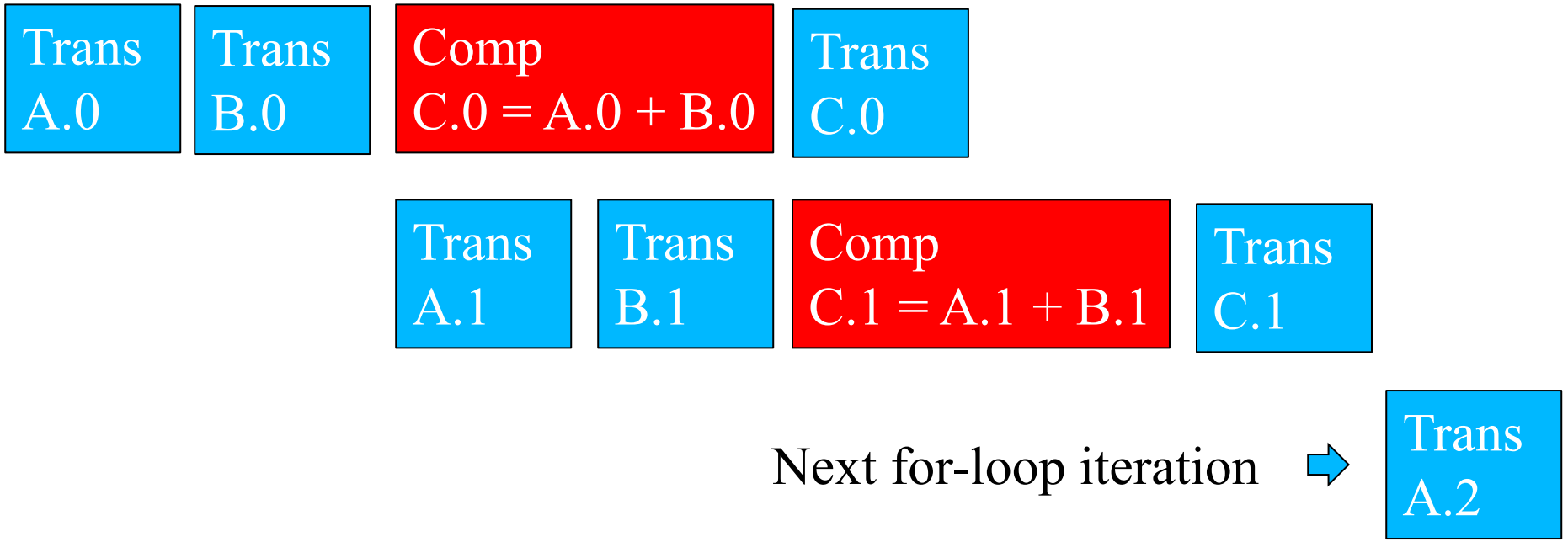


Operations (Kernels, MemCpys)

© David Kirk/NVIDIA and Wen-mei W. Hwu, 2007-2018
ECE408/CS483/ University of Illinois at Urbana-Champaign

Better Overlap with Two Streams

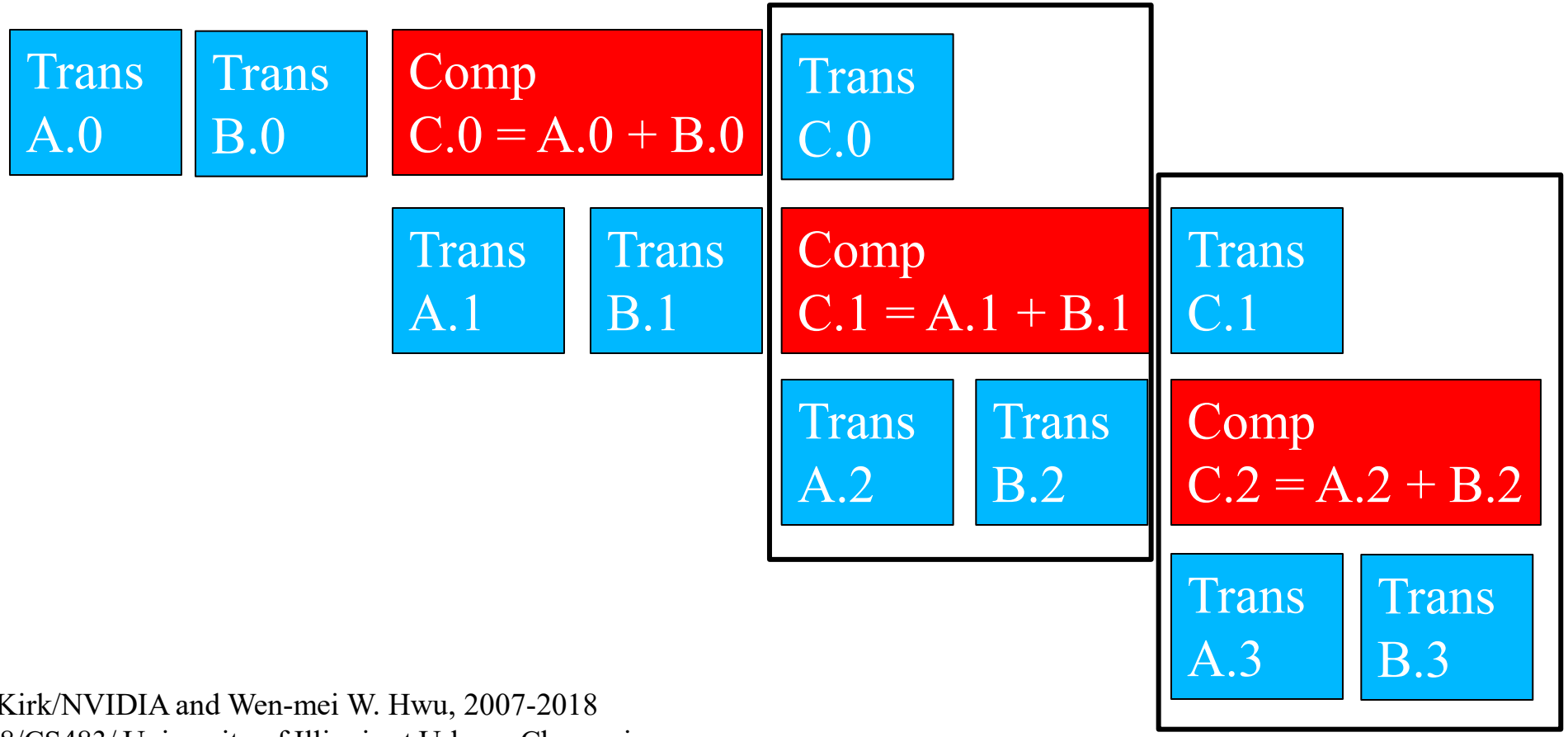
- C.0 no longer blocks A.1 and B.1 in the copy engine queue
- However, C.1 still blocks A.2 and B.2 from the next iteration – PCIe used for only one direction



© David Kirk/NVIDIA and Wen-mei W. Hwu, 2007-2018
ECE408/CS483/ University of Illinois at Urbana-Champaign

Three streams needed for continuous pipelining

- Divide large vectors into segments
- Overlap transfer and compute of adjacent segments



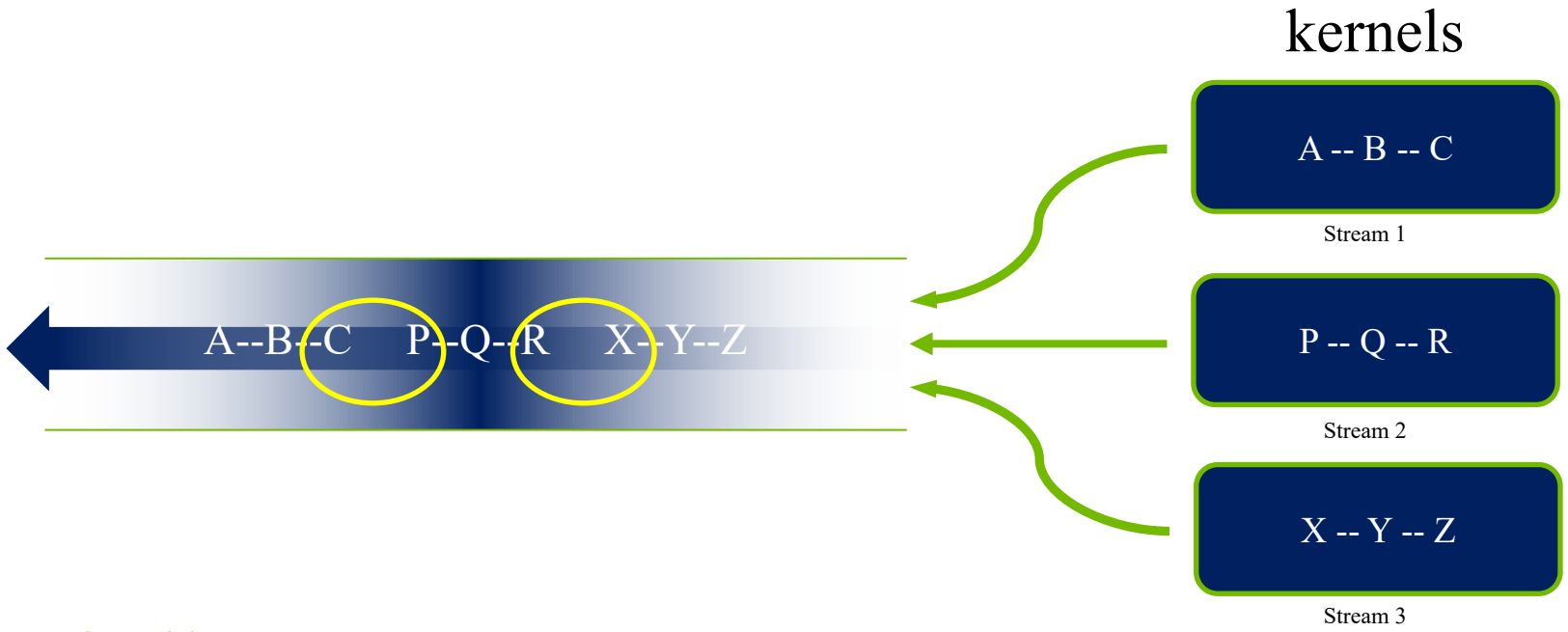
© David Kirk/NVIDIA and Wen-mei W. Hwu, 2007-2018
ECE408/CS483/ University of Illinois at Urbana-Champaign

Hyper Queue

- Provide multiple real stream queues for each engine
- Allow more concurrency by allowing some streams to make progress for an engine while others are blocked

© David Kirk/NVIDIA and Wen-mei W. Hwu, 2007-2018
ECE408/CS483/ University of Illinois at Urbana-Champaign

Fermi (and older) Concurrency

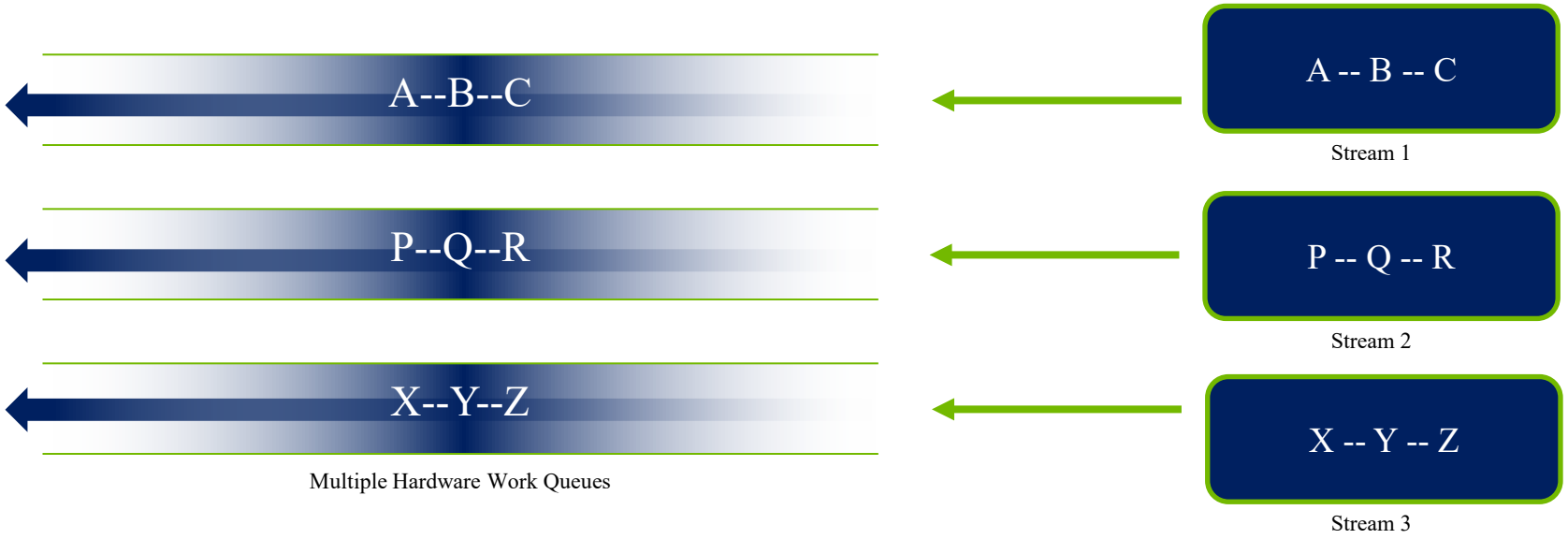


Fermi allows 16-way concurrency

- Up to 16 grids can run at once
- But kernels from CUDA streams multiplex into a single queue
- Overlap only at stream edges

© David Kirk/NVIDIA and Wen-mei W. Hwu, 2007-2018
ECE408/CS483/ University of Illinois at Urbana-Champaign

Kepler Improved Concurrency



Kepler allows 32-way concurrency

- One work queue per stream
- Concurrency at full-stream level
- No inter-stream dependencies

© David Kirk/NVIDIA and Wen-mei W. Hwu, 2007-2018
ECE408/CS483/ University of Illinois at Urbana-Champaign

Smaller Segments Reduce Boundary Effects

How small should segments be?

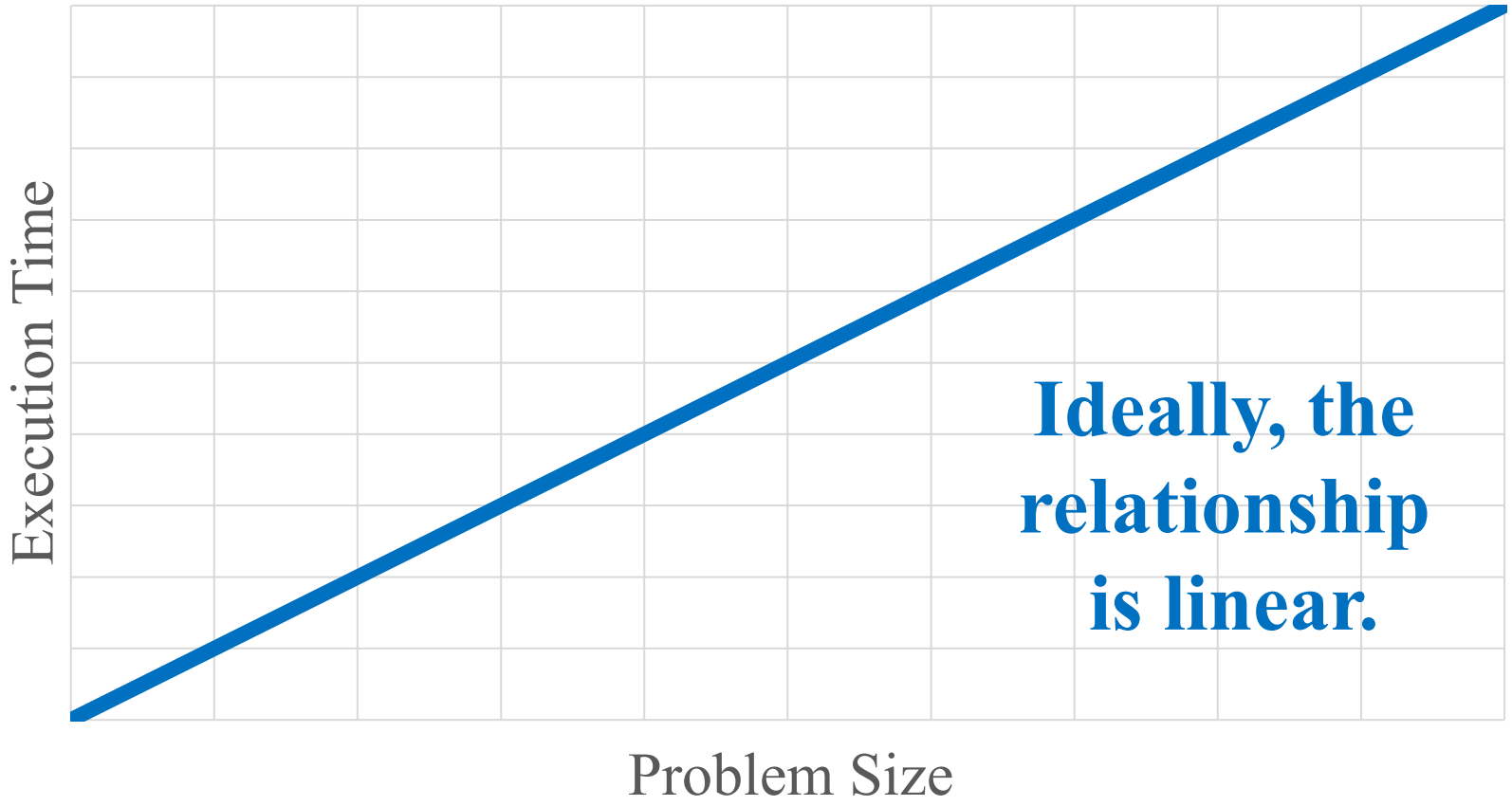
- If we **overlap**
  - **transfer of** segment N's **inputs**,
  - **computation** of segment N – 1, **and**
  - **transfer** of segment N – 2's **results**,
- we **still have non-overlapping work** at the beginning and the end.

So segments should be really small?

© David Kirk/NVIDIA and Wen-mei W. Hwu, 2007-2018
ECE408/CS483/ University of Illinois at Urbana-Champaign

Execution Time is Ideally Linear in Size

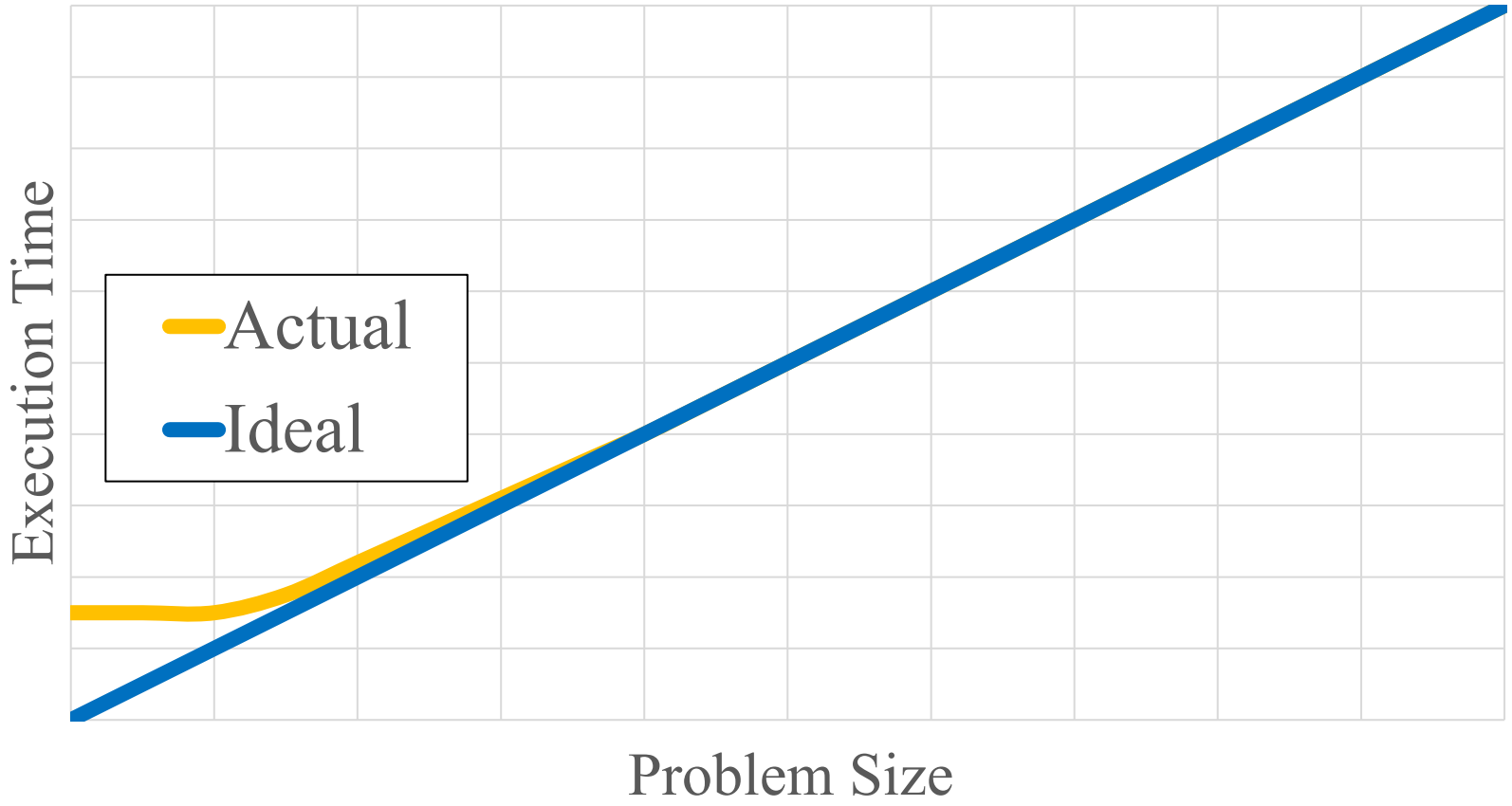
Think about execution time as a function of segment size.



© David Kirk/NVIDIA and Wen-mei W. Hwu, 2007-2018
ECE408/CS483/ University of Illinois at Urbana-Champaign

Execution Time Never Reaches Zero

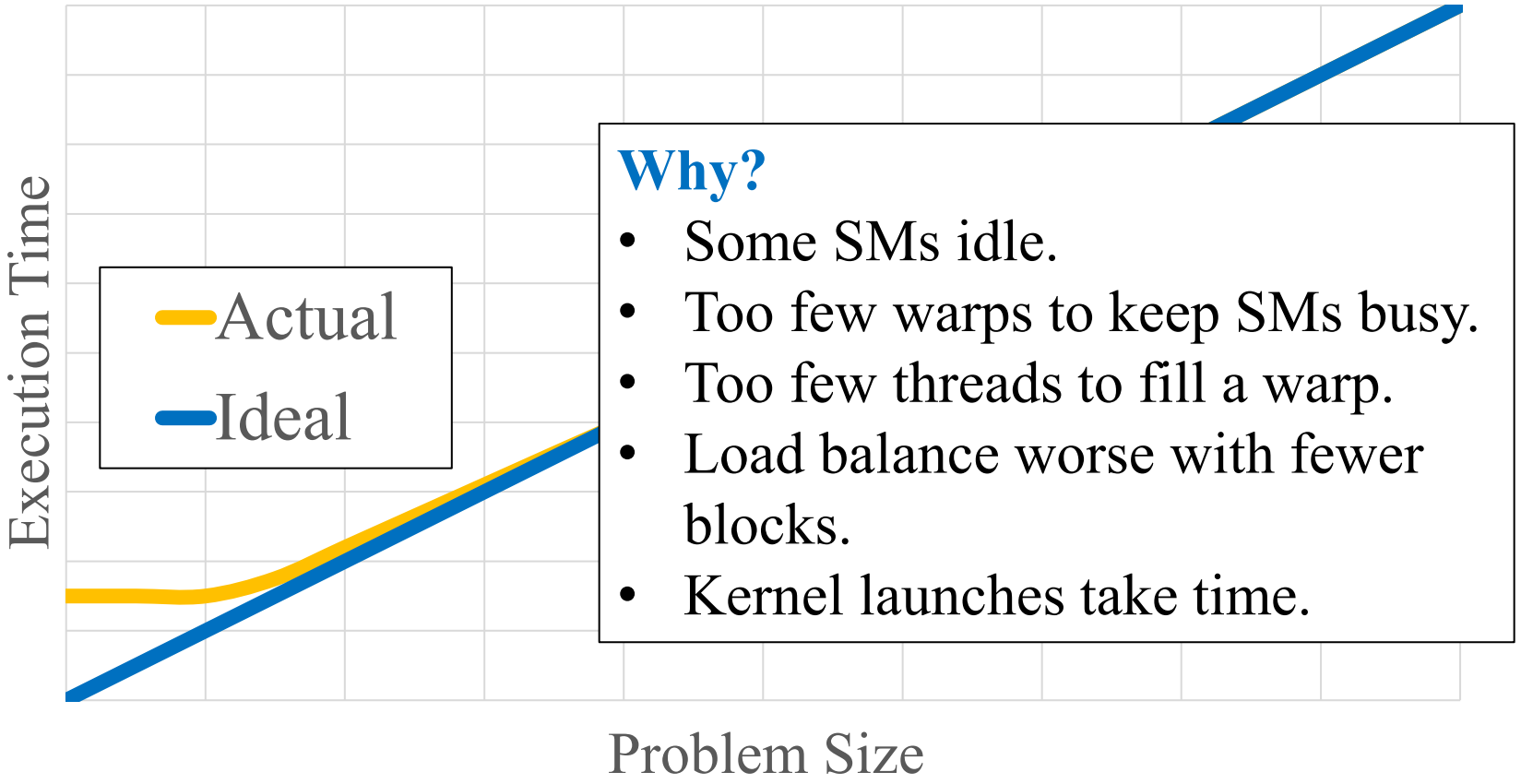
But real execution time has a minimum.



© David Kirk/NVIDIA and Wen-mei W. Hwu, 2007-2018
ECE408/CS483/ University of Illinois at Urbana-Champaign

Execution Time Never Reaches Zero

But real execution time has a minimum.



Why?

- Some SMs idle.
- Too few warps to keep SMs busy.
- Too few threads to fill a warp.
- Load balance worse with fewer blocks.
- Kernel launches take time.

© David Kirk/NVIDIA and Wen-mei W. Hwu, 2007-2018
ECE408/CS483/ University of Illinois at Urbana-Champaign

Use Moderate Segment Size and Device Query

Data transfers

- **have similar non-linearities** for small sizes
- due to startup costs on host and DMA.

So how small should segments be?

Moderately sized.

Best size likely to **depend on GPU**.

© David Kirk/NVIDIA and Wen-mei W. Hwu, 2007-2018
ECE408/CS483/ University of Illinois at Urbana-Champaign

ANY MORE QUESTIONS READ CHAPTER 13

© David Kirk/NVIDIA and Wen-mei W. Hwu, 2007-2018
ECE408/CS483/ University of Illinois at Urbana-Champaign

Problem Solving

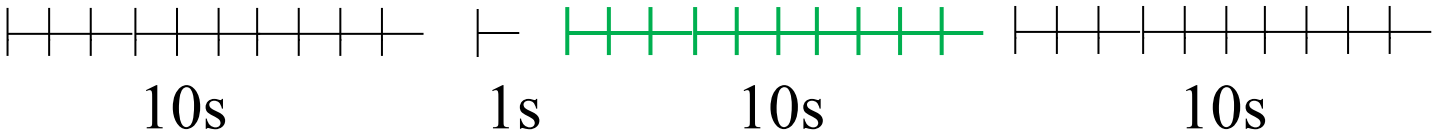
- You are tasked with performing some operations on a very long vector. Upon profiling your kernel code with nsys, you realize that a significant portion of the execution time is taken up by serial memory transfers to and from the device. You decide to use CUDA streams to overlap the memory transfers with computation to decrease the total execution time. There is a constant overhead for starting a kernel computation on the GPU, which is independent of the size of the data that the kernel is processing. (Note that kernel launch cannot overlap with kernel execution.) The following diagram illustrates the timeline for a single stream execution (not to scale).

Input Transfer A (Host to Device)	Kernel Launch Overhead	Compute Time	Output Transfer B (Device to Host)

  The times for each of the section shown above is as follows:
  - Total Input Transfer time: 10 sec
  - Constant Kernel Launch overhead: 1 sec
  - Total Compute time: 10 sec
  - Total Output Transfer Time: 10 sec
- Q: Assume that you have access to 3 hardware streams that enable continuous pipelining. How many segments should the input vector be divided into so as to minimize the total execution time where continuous pipelining is possible?

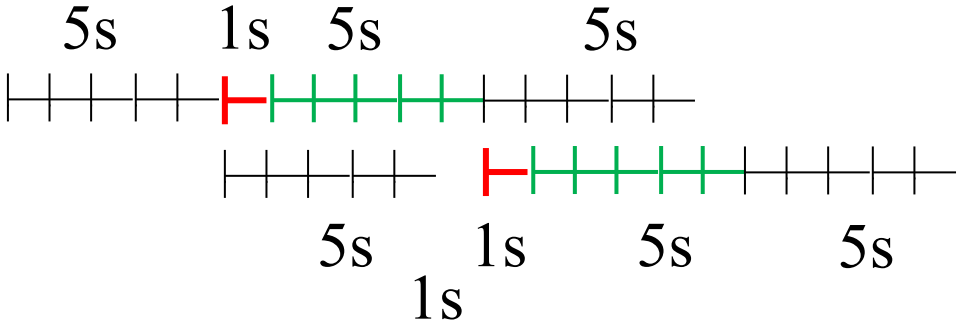
Problem Solving

1 segment
1 stream



31s

2 segments
2 streams



17s